Low Literacy and Educational Inequality in Semi-Urban India: A Case Study of Namauli Village, Greater Noida

Abstract—This paper investigates the persistent issue of low literacy rates among underprivileged children in semi-urban areas of Greater Noida, India. Despite the presence of multiple government programs, including the Right to Education (RTE) Act and Sarva Shiksha Abhiyan, literacy levels within marginalized communities remain critically low. The study adopts a mixed-method approach that combines field surveys, interviews, and geo-tagged documentation of Namauli village to assess the socio-economic, infrastructural, and informational barriers to education. Results indicate that approximately 40.7% of children are not enrolled in school, primarily due to financial hardship, lack of awareness, and parental illiteracy. Nevertheless, over 74% of parents expressed a willingness to educate their children if adequate support and accessibility were provided. The findings suggest that illiteracy in semi-urban India is a multidimensional issue extending beyond poverty to include policy communication gaps and cultural limitations. Sustainable solutions such as community-driven learning centers, awareness programs, and the integration of gamified digital tools are proposed to enhance inclusivity and align with the objectives of the United Nations Sustainable Development Goal 4 (Quality Education).

Index Terms— Low literacy, underprivileged children, semi-urban education, SDG 4, educational inequality, parental awareness, community learning, digital inclusion

INTRODUCTION

Education is universally acknowledged as a fundamental human right and a foundation for social and economic advancement [7]. Despite progress in literacy across India, disparities continue to persist between urban, semi-urban, and rural communities. The national literacy rate of approximately 74% conceals significant inequalities, with semi-urban regions facing persistent challenges related to poverty, limited access, and low parental awareness [8], [14].

Namauli village, situated near Knowledge Park II in Greater Noida, exemplifies this divide. The village lies adjacent to major educational and industrial hubs yet exhibits characteristics of rural educational deprivation. Although schools and institutions are physically accessible, financial instability, parental illiteracy, and misconceptions about schooling expenses prevent many children—especially girls—from attending or continuing education [3], [8].

Government initiatives such as the Right to Education (RTE) Act (2009), Sarva Shiksha Abhiyan, and the Mid-Day Meal Scheme have significantly improved enrollment rates at the national level [2], [6]. However, gaps remain in local implementation and community awareness, limiting their effectiveness [10], [12]. Addressing this challenge requires a multi-faceted strategy that combines policy outreach, financial assistance, and innovative learning interventions to improve inclusion and retention.

As illustrated in Fig. 1, Namauli village's geographical location near Greater Noida highlights the paradox between infrastructural proximity and educational inequality. This setting provides a relevant context for analyzing barriers to literacy and identifying sustainable, community-centered solutions.

Fig. 1. Location of Namauli village near Knowledge Park II, Greater Noida, illustrating its semi-urban context and proximity to educational infrastructure [15].

LITERATURE REVIEW

Previous research has identified multiple interrelated factors contributing to low literacy among underprivileged children. Most studies agree that the challenge extends beyond access to schooling, encompassing economic deprivation, gender inequality, infrastructure deficits, and policy implementation failures.

Economic and Social Barriers

Poverty continues to be a key determinant of low literacy levels across developing regions. Studies by Javed et al. [3] and Roy [8] highlight that economic hardship often forces children into income-generating activities, reducing school attendance and increasing dropout rates. In such households, immediate survival outweighs long-term educational investment. Gender bias further compounds this issue, as traditional norms prioritize male education while limiting opportunities for girls [14].

Infrastructure and Resource Deficits

Infrastructural inadequacies significantly affect the quality and continuity of education. Research from Trichy and Delhi slums indicates that over 80% of rural schools lack proper classrooms, sanitation facilities, or libraries, directly impacting student motivation and attendance [7], [10]. Even when schools exist, overcrowded classrooms, irregular teacher availability, and poor maintenance result in low learning outcomes. These studies underscore that infrastructure alone is insufficient without parallel efforts in teacher quality and student engagement.

Awareness and policy Implementation Gaps

Despite the introduction of government initiatives such as the Right to Education (RTE) Act, Sarva Shiksha Abhiyan, and the Mid-Day Meal Scheme, many families remain unaware of their provisions [1], [2], [6]. Suman [7] and Saha et al. [14] found that weak community outreach, misuse of funds, and inconsistent monitoring hinder the success of these programs. Parents often perceive “free education” as involving hidden costs such as uniforms and transportation, creating mistrust and low participation.

Pedagogical Innovation and Inclusive Learning

Recent discourse emphasizes the need to move beyond conventional teaching methods. Murray [4] argues that literacy should encompass multiple dimensions—digital, emotional, and visual—rather than being confined to reading and writing. Similarly, inclusive education research advocates flexible, learner-centered approaches and the use of local resources to sustain engagement among children from low-literacy families [11], [13]. Gamified and digital tools are increasingly being recognized for their potential to enhance motivation and bridge learning gaps in marginalized communities.

Identified Research Gap

While extensive work has been conducted on rural and urban education systems, semi-urban areas like Namauli remain underexplored. These communities face a dual burden—limited rural-level support and urban-level cost pressures. Furthermore, few studies address how information asymmetry and community awareness specifically influence literacy outcomes. The present research addresses this gap through empirical fieldwork in Namauli village, combining survey data, interviews, and geo-tagged documentation to develop context-specific interventions.

METHODOLOGY

Research Design

This study adopts a mixed-method approach, integrating both quantitative and qualitative techniques to examine the factors influencing literacy levels among underprivileged children in semi-urban regions. The research combines field-based surveys, structured interviews, and geo-tagged observations conducted in Namauli village, Greater Noida. This design enables a comprehensive understanding of both numerical trends and contextual realities [3], [7].

Study Area

The study was carried out in Namauli village, located near Knowledge Park II in Greater Noida, Uttar Pradesh. The area was selected for its transitional semi-urban character, where access to schools and digital infrastructure coexists with poverty, low literacy, and minimal policy awareness. The village serves as a representative site for analyzing the educational challenges of communities positioned between rural and urban settings.

Participants and Sampling

A total of 32 respondents participated in the survey, comprising parents, local residents, and NGO workers. The sample was selected through purposive sampling, focusing on households with at least one school-aged child. The demographic distribution included both male and female participants from low- to middle-income families, ensuring representativeness across social and economic backgrounds.

The demographic distribution of respondents, including age, gender, and family composition, is summarized in Table I.

TABLE I Demographic Profile of Respondents

Data Collection Tools

Google Form Survey: A structured questionnaire was designed to collect quantitative data on demographics, income, school enrollment, awareness of government schemes, and educational preferences.

Interviews and Field Notes: Semi-structured interviews were conducted with parents and local teachers to obtain qualitative insights into cultural and motivational barriers.

Geo-tagged Field Photographs: Images of schools, community areas, and children’s study environments were captured to document local conditions visually.

Secondary Sources: Government databases, journal articles, and NGO reports were reviewed to triangulate findings and support analysis [8], [10].

Data Analysis Procedure

Quantitative data from the survey were analyzed using descriptive statistics (frequency and percentage distributions), while qualitative responses were coded thematically into categories such as economic barriers, infrastructure gaps, and awareness deficits. The results were interpreted collectively to identify patterns and relationships influencing literacy outcomes.

The quantitative results obtained from the field survey are summarized in Fig. 3, which highlights the major factors influencing school enrollment and literacy within the Namauli community.

Fig. 3. Summary of key survey findings highlighting financial, informational, and social barriers to children’s education in Namauli village.

Workflow of the Study

The overall research workflow is illustrated in Fig. 2, which outlines the sequential stages of the study — from data collection to intervention development.

Fig. 2. Workflow of the study showing sequential stages

The workflow begins with survey and field data collection, followed by data organization and cleaning using descriptive methods. Subsequently, quantitative trends and qualitative themes were integrated to identify literacy barriers. The final stage involved formulating targeted recommendations—including awareness programs, NGO–government collaboration models, and digital literacy initiatives—to enhance inclusivity and educational participation in semi-urban communities.

RESULT AND ANALYSIS

Enrollment and Attendance

The field survey revealed that 40.7% of children in Namauli village were not enrolled in school, while 59.3% were attending regularly. The primary reasons cited for non-enrollment were financial hardship (65%), distance to school (25%), and household responsibilities (10%). Among enrolled students, irregular attendance was observed during seasonal labor periods, reflecting the financial instability of families.

Despite these challenges, 85.7% of respondents agreed that education should be prioritized over other expenses. This indicates that the barrier is not a lack of interest but rather limited access and affordability.

Fig. 4. Enrollment status of children in Namauli village showing the percentage of enrolled vs. non-enrolled respondents.

Awareness and Access to Government Schemes

Awareness of educational schemes was found to be inconsistent. Only 56% of respondents recognized major government initiatives such as the Right to Education (RTE) Act, Mid-Day Meal Scheme, and Pradhan Mantri Awas Yojana. However, awareness specifically related to education was markedly lower.

Interviews revealed that some parents believed these schemes were intended only for urban schools, while others expressed confusion about “free education” and possible hidden costs (uniforms, books, or transportation). This demonstrates a significant communication gap between policy and practice [10], [14].

Fig. 5. Awareness levels of government schemes among respondents in Namauli village.

Socio-Economic Context

The socio-economic analysis showed that 60% of households lived below the poverty line. About 32.1% depended on salaried jobs, 25% on self-employment, and another 25% on government assistance. More than half (53.6%) of families reported that education formed the largest portion of their monthly expenses, demonstrating a strong aspirational mindset despite limited income.

Half of the respondents (50%) saved money regularly, though most savings were below ■500 per month, reflecting financial vulnerability. Encouragingly, 74.1% of parents stated they would educate their children if provided with financial or logistical support.

Fig. 6. Socio-economic composition of surveyed households showing income sources and savings behavior.

Gender and Educational Motivation

Gender inequality emerged as a persistent factor influencing literacy. While both parents valued schooling, girls’ education was often deprioritized due to safety concerns and traditional social expectations. Qualitative interviews revealed that girls were frequently tasked with domestic duties, while boys were encouraged to pursue schooling.

Interestingly, several children demonstrated intrinsic motivation for learning — one respondent described a young boy who attended school in the morning and practiced mehndi art in the evening to support his family, reflecting creativity and determination.

Interpretation of Findings

The analysis confirms that low literacy in semi-urban communities like Namauli arises not from apathy but from a combination of economic stress, poor policy awareness, and gendered social norms. While financial limitations restrict school continuity, information poverty further prevents families from leveraging available educational schemes. However, the expressed willingness of parents to support education—if aided through awareness campaigns and financial assistance—demonstrates the potential for community-led improvement.

To contextualize local findings, the identified barriers were compared with national-level literacy data from ASER (2023), as presented in Table II.

TABLE II Comparative Summary of Literacy Barriers: Namauli vs. National Data

Comparative Insights

To contextualize the findings, the results from Namauli were compared with national-level literacy data from the Annual Status of Education Report (ASER, 2023). While the national average for school enrollment amongchildren aged 6–14 stands at 96%, Namauli's effective enrollment rate of 59.3% reflects the persisting educational gap between rural and semi-urban communities. The 44% lack of awareness regarding government schemes also exceeds the national average of 31%, underscoring the role of localized communication barriers in limiting access to education.

DISCUSSION

The field data from Namauli village reveal that low literacy among underprivileged children arises not from apathy but from a combination of economic, informational, and cultural barriers. The results reinforce earlier findings by Javed et al. [3] and Roy [8], who noted that financial instability and poverty remain key deterrents to school attendance. However, this study expands the discussion by highlighting the role of information poverty—families' limited awareness of educational schemes—even in semi-urban areas that are geographically close to schools.

A. Economic Constraints

More than half of the surveyed households live below the poverty line, compelling children to supplement family income. Although 85.7 % of parents considered education a priority, economic insecurity caused irregular attendance and early dropouts. This confirms the view that poverty is not merely a lack of income but a barrier to educational continuity [8], [9]. Micro-level interventions such as stipends, transport

aid, and free learning materials could immediately mitigate these constraints.

B. Awareness and Implementation Gaps

The study found that 44 % of parents were unaware of the RTE Act or the Mid-Day Meal Scheme, illustrating a clear communication breakdown between policy and beneficiaries. Consistent with Suman [7] and Saha et al. [14], weak outreach and poor administrative follow-through hinder otherwise effective national programs. Strengthening local dissemination—through community volunteers, school open-house meetings, and NGO-led campaigns—could significantly enhance participation.

C. Gender and Cultural Dimensions

Gender bias remains a persistent barrier. Families tend to prioritize sons' education while assigning domestic responsibilities to daughters. Similar observations have been reported in rural India [14], [13]. Encouraging female literacy requires culturally sensitive strategies such as women-led learning groups, parental counseling, and ensuring safe transportation for girls.

D. Role of Pedagogy and Technology

Traditional literacy programs focus narrowly on reading and writing. Murray [4] argues for embracing "multiple literacies" that integrate digital, emotional, and visual communication. Aligning with this perspective, the present study recommends the introduction of gamified digital tools and low-bandwidth mobile apps (e.g., Google Read Along, Kahoot!) to improve engagement and retention. These tools, when used in community centers, can make learning interactive and inclusive even for first-generation learners.

E. Towards Sustainable Interventions

The convergence of economic, informational, and cultural barriers suggests the need for a multi-stakeholder framework involving schools, NGOs, and local administrations. Establishing evening community learning pods, integrating vocational and digital training, and conducting localized awareness drives can produce sustainable literacy growth. Such initiatives align with Sustainable Development Goal 4 (Quality Education) by promoting equitable access and lifelong learning opportunities.

The proposed interventions and their potential outcomes are summarized in Table III, emphasizing collaborative approaches for sustainable educational development.

TABLE III Proposed Interventions for Literacy Improvement in Semi-Urban Communities

F Community Participation and Governance

Effective literacy improvement in semi-urban settings requires the active participation of local governing bodies and community leaders. Establishing partnerships between village panchayats, school management committees, and non-governmental organizations can help bridge administrative and communication gaps. Local volunteers can act as education ambassadors, organizing awareness drives, mentoring children, and promoting attendance through door-to-door campaigns. Empowering communities through shared accountability not only enhances policy implementation but also ensures long-term sustainability of literacy initiatives.

CONCLUSION AND FUTURE WORK

The study concludes that low literacy among underprivileged children in semi-urban regions such as Namauli village, Greater Noida, results primarily from economic hardship, low parental awareness, and socio-cultural limitations, rather than a lack of interest in education. Although 40.7 % of children remain unenrolled, the strong willingness of parents to educate their children—if supported financially and institutionally—indicates significant untapped potential. Existing government initiatives like the RTE Act, Sarva Shiksha Abhiyan, and Mid-Day Meal Scheme have improved enrollment nationally, but their local impact remains limited by weak outreach and policy implementation. Addressing these barriers requires community participation, targeted financial aid, improved policy communication, and technology-driven learning interventions. Collectively, these measures align with the United Nations Sustainable Development Goal 4 (Quality Education) by promoting inclusive and equitable access to education.

Future work should extend this analysis to other semi-urban and peri-rural areas to compare regional disparities and develop scalable intervention models. Longitudinal studies could assess the sustainability of awareness campaigns, NGO–government partnerships, and digital literacy programs in improving learning outcomes. Integrating gamified mobile applications, low-cost digital platforms, and AI-based adaptive learning systems could further enhance engagement and data-driven monitoring of literacy improvement over time.

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